

NAME : CHETHAN BJ

SRN : PES1UG24AM426

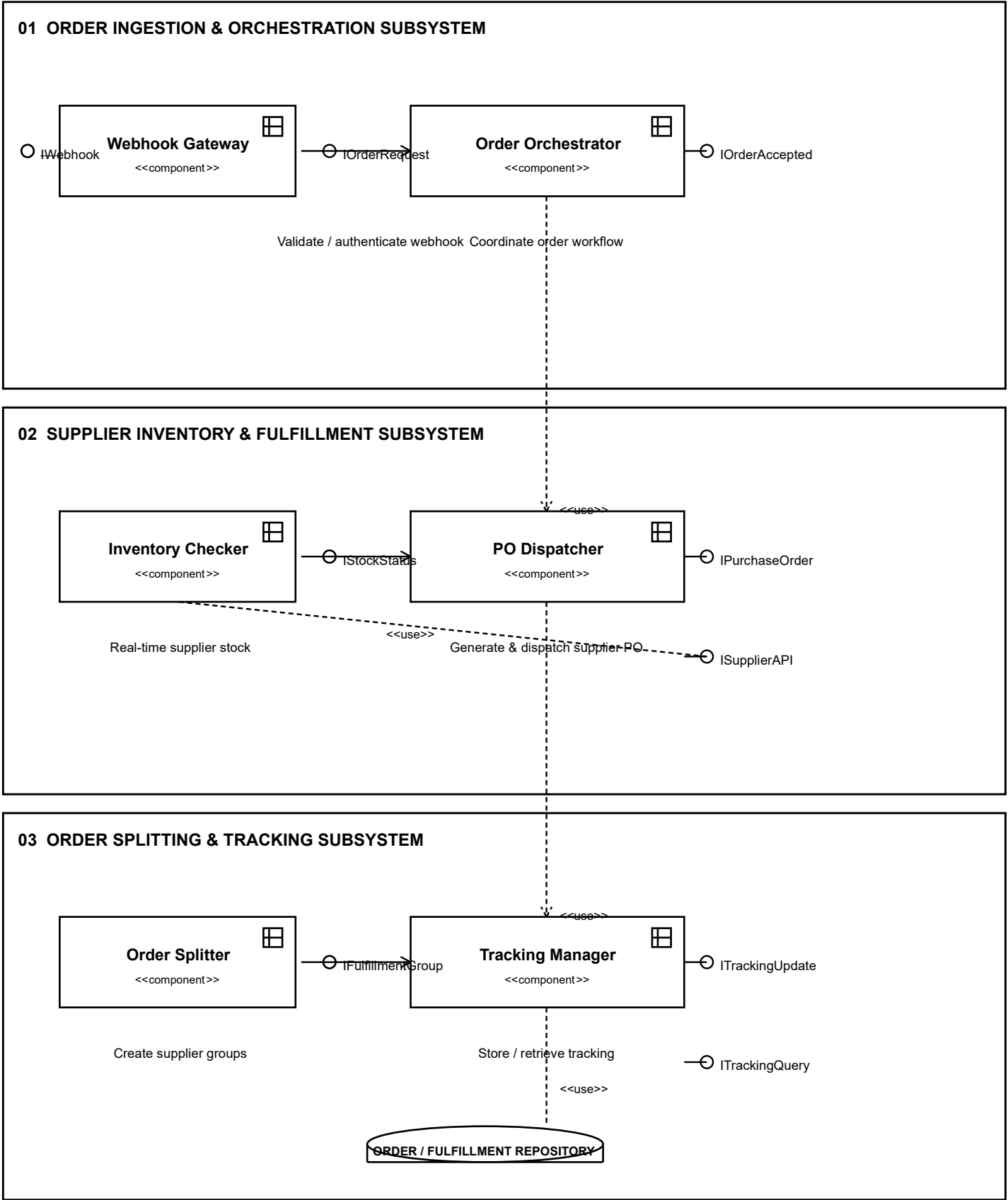
SECTION : H

LAB : 03

PROBLEM STATEMENT #38

DROPSHIPPING INVENTORY & ORDER ORCHESTRATOR

COMPONENT DIAGRAM



Architecture Selection:

Microservices Architecture

We chose the Microservices Architecture for the Dropshipping Inventory & Order Orchestrator. The first reason for selecting this architecture is clear separation of responsibilities. The system performs several distinct tasks, including receiving storefront webhooks, validating orders, checking supplier inventory, splitting multi-vendor orders, dispatching supplier purchase orders, and tracking shipments. By separating these responsibilities into independently deployable services, the system becomes easier to understand, maintain, test, and extend.

The second reason is the system's dependence on multiple third-party supplier APIs. Supplier integrations may have different response formats, availability, and performance characteristics. With microservices, supplier inventory checking and purchase-order dispatch can be isolated behind clear interfaces, allowing one supplier integration to change without requiring changes to the entire application. The Order Orchestrator coordinates these services and maintains the overall order-processing flow.

A major security advantage of the selected architecture is isolation of external integrations. The Webhook Gateway can authenticate and validate incoming storefront requests before forwarding trusted data to the Order Orchestrator. Supplier API credentials can be kept inside the appropriate integration service, while sensitive order information can be controlled through dedicated data-access components. This limits unnecessary exposure of customer and supplier information.

A key performance benefit is independent scaling and asynchronous processing. During peak order periods, webhook ingestion and order orchestration can scale separately from supplier stock verification and tracking updates. Supplier stock checks can be processed concurrently, and tracking updates can be handled without blocking new incoming orders. This helps the system maintain responsiveness while processing multiple supplier interactions.

Therefore, the Microservices Architecture provides a modular, scalable, secure, and maintainable solution for the Dropshipping Inventory & Order Orchestrator. It supports the required separation between webhook handling, order coordination, supplier inventory, fulfillment, and shipment tracking while addressing the important security and performance requirements of the scenario.